# Lecture 8: Definite Matrices and the Singular Value Decomposition

### Through a Flat-to-3D Filter Problem

AI 501: Mathematics for Artificial Intelligence

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**Abstract**

Eigen-decomposition only works for square matrices. The singular value decomposition (SVD) works for *every* matrix. The idea in one sentence: every linear map rotates, stretches along perpendicular axes, and rotates again, so it turns a circle into an ellipse. We carry one example through the notes, a filter that lifts a flat picture into 3D, and use the SVD to see what a matrix does, how many dimensions it really uses, and how to replace it by a simpler matrix. The order of the notes is: the problem, a small tool (definite matrices), the SVD and its geometry, how to compute it, and two applications (rank and approximation).

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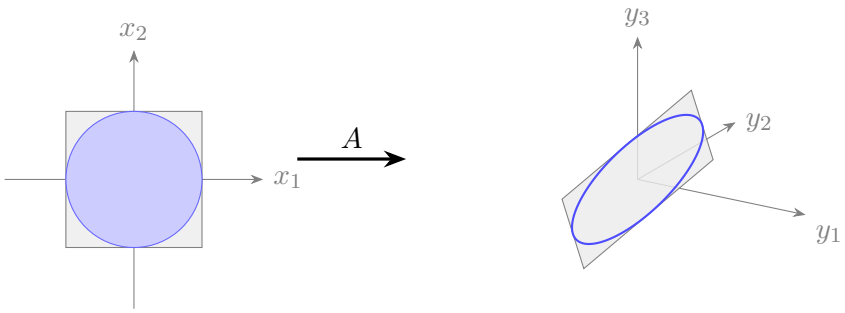
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## 1 One problem: a filter that lifts a picture into 3D

A graphics program takes a flat picture and lifts it onto a tilted sheet in 3D. A pixel $x = (x_1, x_2)^T \in \mathbf{R}^2$ is sent to the point $Ax \in \mathbf{R}^3$, where

$$A = \begin{bmatrix} 1 & -0.8 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \in \mathbf{R}^{3\times 2}.$$

To see what the filter does, we feed it a test picture: a round logo of radius 1 on a square canvas. Figure 1 shows the result. The square becomes a parallelogram, and the round logo becomes a




canvas $\mathbf{R}^2$ image in $\mathbf{R}^3$: a tilted oval on a flat sheet

Figure 1: The lifting filter $A$ applied to a test image: a round logo on a square canvas. The image of the whole canvas is a flat parallelogram inside $\mathbf{R}^3$, and the logo becomes an oval on it.

tilted oval, *lying on one flat sheet* in 3D. Every output satisfies $y_3 = y_1 + 0.8\, y_2$, so although the output lives in $\mathbf{R}^3$, it only fills a two-dimensional sheet.

Think of a matrix as a machine that moves points around. We would like a simple description of what the machine does. In Lecture 7 we found one for symmetric matrices: there are perpendicular directions along which the matrix *only stretches*. For our filter that tool is not available, because an eigenvector needs $Aq = \lambda q$, so $Aq$ and $q$ must live in the same space, but here $q \in \mathbf{R}^2$ and $Aq \in \mathbf{R}^3$. (Even for square matrices, the eigenvectors of a non-symmetric matrix need not be perpendicular, and a rotation has no real eigenvalues at all.)

The SVD repairs this by allowing *different* directions on the input side and on the output side. Our questions are:

1. What does *every* matrix do to a circle, and which perpendicular directions does it stretch? (Sections 3–4)
2. How do we find those directions? This needs one small tool, definite matrices. (Sections 2 and 5)
3. How many dimensions does the output really use, and can we replace $A$ by something simpler? (Sections 6–7)

## 2 A small tool: positive and negative definite matrices

### 2.1 What the quadratic form measures

For a square matrix $A \in \mathbf{R}^{n\times n}$ and a vector $x \in \mathbf{R}^n$, the number

$$x^T A x = x \cdot (Ax)$$

is called a *quadratic form*. It is a single number (a $1 \times n$ row times an $n \times n$ matrix times an $n \times 1$ column), and it has a simple meaning: it is the dot product of $x$ with its own image $Ax$. If $A$ keeps $x$ pointing roughly the same way, the dot product is positive. If $A$ turns $x$ around to point roughly the opposite way, it is negative. So the sign of $x^T A x$ tells us whether $A$ "agrees with" or "fights" the vector $x$. We classify matrices by this sign.

**Definition.** $A \in \mathbf{R}^{n\times n}$ is

- *positive semi-definite* (PSD) if $x^T A x \geq 0$ for all $x$;
- *positive definite* (PD) if $x^T A x > 0$ for all $x \neq 0$;
- *negative semi-definite* (NSD) / *negative definite* (ND) if the same holds with $\leq 0$ / $< 0$;
- *indefinite* if $x^T A x$ takes both signs.

(The strict versions need $x \neq 0$, since $x^T Ax = 0$ at $x = 0$ for every matrix.)

**In words.** A PSD matrix never turns a vector by more than a right angle: $x \cdot Ax \geq 0$ means the angle between $x$ and $Ax$ is at most 90° (Figure 2). A PD matrix keeps it strictly below 90°. An indefinite matrix turns some vectors forward and others backwards.

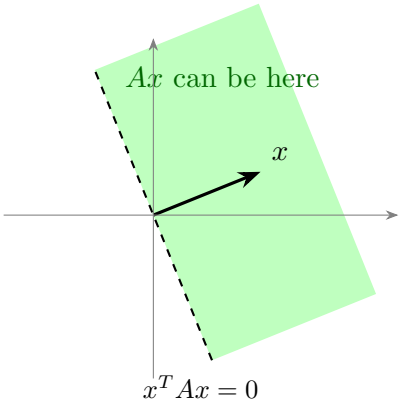



Figure 2: For a fixed $x$, a PSD matrix sends $x$ somewhere in the shaded half-plane. The dashed line is perpendicular to $x$: on it $x^T Ax = 0$. For a PD matrix the dashed boundary is excluded.

## 2.2 Eigenvalues decide the sign

How can we tell, without testing every $x$, which kind of matrix we have? For a *symmetric* matrix (the case we need), the eigenvalues tell us. The reason is worth seeing. Write $A = Q\Lambda Q^T$ (Lecture 7) and describe $x$ in the eigenvector basis, $x = \sum_i c_i q_i$ with $c = Q^T x$. In that basis, $A$ simply stretches the $i$th coordinate by $\lambda_i$, and

$$x^T Ax = \sum_{i=1}^{n} \lambda_i c_i^2.$$

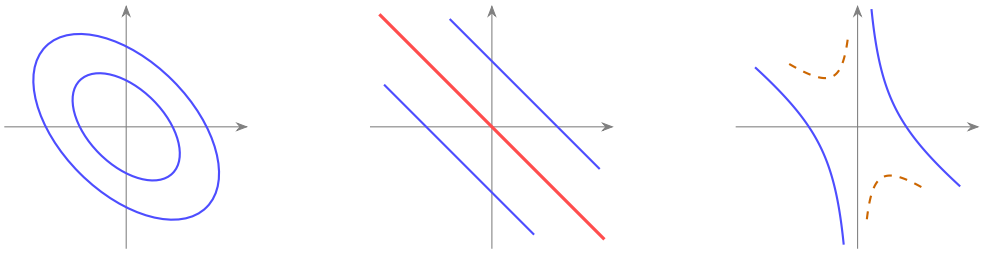
**In words:** the quadratic form is a sum of squares, each multiplied by an eigenvalue. A sum of squares weighted by positive numbers cannot be negative, and one weighted by mixed signs can be either. So the signs of the eigenvalues decide everything:

**Key fact.** A symmetric matrix is PD $\iff$ all $\lambda_i > 0$; PSD $\iff$ all $\lambda_i \geq 0$; ND $\iff$ all $\lambda_i < 0$; NSD $\iff$ all $\lambda_i \leq 0$; and it is indefinite if the signs are mixed.

(For the "only if" direction, take $x$ to be a unit eigenvector: then $x^T Ax = \lambda$.) Figure 3 shows the three pictures through the curves $x^T Ax = \text{const}$: closed ellipses when all weights are positive, a stretched-out strip when one weight is zero, and a saddle when the signs differ.

**Examples** (the matrices of Lecture 7). $S = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$ has $\lambda = 3, 1$, both positive, so it is **PD** (directly: $x^T Sx = x_1^2 + x_2^2 + (x_1 + x_2)^2 > 0$). $F = \frac{1}{2}\left[\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right]$ has $\lambda = 1, 0$, so it is **PSD but not PD**: $x^T Fx = \frac{1}{2}(x_1 + x_2)^2$ is never negative, but it is 0 at $x = (1, -1)^T$, the eigenvector of $\lambda = 0$. $M = \left[\begin{smallmatrix} 3 & 2 \\ 2 & 0 \end{smallmatrix}\right]$ has $\lambda = 4, -1$, so it is **indefinite**: $x = (1, 0)^T$ gives $+3$ while $x = (1, -2)^T$ gives $-5$.

**Check yourself.** Is $\left[\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right]$ positive definite?
*Answer.* Its trace is 2 and its determinant is $-3$, so $\lambda = 3, -1$: indefinite, not PD.



PD ($\lambda = 3, 1$)
closed curves

PSD ($\lambda = 1, 0$)
parallel lines; zero along the red line

indefinite ($\lambda = 4, -1$)
hyperbolas (a saddle)

Figure 3: Level curves $x^T Ax = c$ for the matrices $S = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$ (PD), $F = \frac{1}{2}\left[\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right]$ (PSD) and $M = \left[\begin{smallmatrix} 3 & 2 \\ 2 & 0 \end{smallmatrix}\right]$ (indefinite) from Lecture 7. For PD the curves are ellipses whose axes lie along the eigenvectors, with half-lengths $\sqrt{c/\lambda_i}$. A zero eigenvalue stretches the ellipse to infinity along its eigenvector, and mixed signs open it into a hyperbola (solid: positive values, dashed: negative values).

### 2.3 Gram matrices measure stretching

Why do we care? Take any matrix $A \in \mathbf{R}^{m \times n}$, not necessarily square. We cannot ask for the eigenvalues of $A$, but we can build two square symmetric matrices from it, $A^T A \in \mathbf{R}^{n \times n}$ and $AA^T \in \mathbf{R}^{m \times m}$. Look at the quadratic form of the first one:

$$x^T (A^T A) x = (Ax)^T (Ax) = \|Ax\|^2.$$

**In words:** the quadratic form of $A^T A$ is the *squared length of the image*. So $A^T A$ knows how much $A$ stretches each vector $x$. This is the key observation behind the SVD (Section 5).

> **Key fact.** For every $A$, the matrices $A^T A$ and $AA^T$ are symmetric and positive semi-definite, so all their eigenvalues are real and $\geq 0$. Moreover $A^T A$ is PD exactly when the columns of $A$ are linearly independent.

*Why.* A squared length $\|Ax\|^2$ is never negative, and it is 0 only if $Ax = 0$, which for independent columns happens only at $x = 0$. The same argument applied to $A^T$ gives $AA^T$. Because the eigenvalues are $\geq 0$, their square roots exist: these will be the *singular values*.

**For the filter.** $A^T A = \left[\begin{smallmatrix} 2 & -0.8 \\ -0.8 & 1.64 \end{smallmatrix}\right] \in \mathbf{R}^{2 \times 2}$ is PD because the two columns of $A$ are independent, while $AA^T \in \mathbf{R}^{3 \times 3}$ is only PSD (the sheet is flat, so one direction in $\mathbf{R}^3$ is never reached).

**Check yourself.** If $A \in \mathbf{R}^{5 \times 3}$ has rank 3, which of $A^T A$, $AA^T$ is PD? How many zero eigenvalues does the other have?
*Answer.* $A^T A$ ($3 \times 3$, rank 3) is PD. $AA^T$ ($5 \times 5$) has rank 3, so it has $5 - 3 = 2$ zero eigenvalues.

## 3 The singular value decomposition

Here is the main idea, in words. However complicated a matrix looks, it does three simple things in a row: it *rotates* the input, *stretches* it along perpendicular axes, and *rotates* the result. The SVD writes this down as a factorisation.

**Theorem (SVD).** Every $A \in \mathbf{R}^{m\times n}$ can be written

$$\boxed{A = U\,\Sigma\,V^T}$$

where $U \in \mathbf{R}^{m\times m}$ and $V \in \mathbf{R}^{n\times n}$ are orthogonal matrices (their columns are perpendicular unit vectors) and $\Sigma \in \mathbf{R}^{m\times n}$ is diagonal with entries $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$, called the *singular values*.

The columns $u_i$ of $U$ are the *left singular vectors* and live in the output space $\mathbf{R}^m$; the columns $v_i$ of $V$ are the *right singular vectors* and live in the input space $\mathbf{R}^n$. Because $\Sigma$ has the shape of $A$, there are $\min(m, n)$ singular values, and $\Sigma$ has extra zero columns if $A$ is wide and extra zero rows if $A$ is tall. For our filter ($m = 3$, $n = 2$) the shapes are

$$\underbrace{A}_{3\times 2} = \underbrace{U}_{3\times 3}\ \underbrace{\Sigma}_{3\times 2}\ \underbrace{V^T}_{2\times 2}.$$

It is always worth tracking these sizes: $V^T$ acts on inputs in $\mathbf{R}^n$, $\Sigma$ converts from $\mathbf{R}^n$ to $\mathbf{R}^m$, and $U$ acts on outputs in $\mathbf{R}^m$.

**The one sentence to remember.** Multiplying $A = U\Sigma V^T$ on the right by $V$ gives $AV = U\Sigma$, and reading this column by column,

$$Av_i = \sigma_i u_i.$$

**In words:** *$A$ sends each right singular vector $v_i$ to the left singular vector $u_i$, stretched by the factor $\sigma_i$.* The $v_i$ are special input directions; along each of them, $A$ does nothing but stretch (and move to the direction $u_i$ in the output space). Everything else in the SVD follows from this.

**Compact form.** If only $r$ singular values are nonzero, the remaining columns of $U$ and $V$ only ever multiply zeros and can be dropped: $A = U_r\Sigma_r V_r^T$ with shapes $(m \times r)(r \times r)(r \times n)$.

**Check yourself.** For $A \in \mathbf{R}^{5\times 3}$, give the shapes of $U, \Sigma, V^T$ and the number of zero rows of $\Sigma$.
*Answer.* $5 \times 5$, $5 \times 3$, $3 \times 3$. There are 3 singular values and the last $5 - 3 = 2$ rows of $\Sigma$ are zero.

# 4 Geometric interpretation: rotate, stretch, rotate

## 4.1 From a circle to an ellipse

Picture the unit circle drawn on the input plane and follow it through $Ax = U\Sigma V^T x$, reading from right to left:

1. $V^T$ **rotates** the circle (more precisely, it re-describes every point using the perpendicular directions $v_1, \ldots, v_n$ as axes). A rotated circle is the same circle: lengths and angles do not change.
2. $\Sigma$ **stretches** the $i$th axis by $\sigma_i$. The circle becomes an ellipse. This step also changes the dimension from $n$ to $m$: if $m < n$ the extra coordinates are dropped, and if $m > n$ zeros are appended.
3. $U$ **rotates** the ellipse onto the directions $u_1, \ldots, u_m$ of the output space.

Altogether, *a circle becomes a tilted ellipse*, and the SVD tells us exactly which one:

**Key fact (circles become ellipses).** The image of the unit ball is an ellipsoid whose semi-axes are $\sigma_1 u_1,\ \sigma_2 u_2, \ldots, \sigma_r u_r$. So the *lengths* of the axes are the singular values, their *directions* are the left singular vectors, and $\sigma_1$ is the largest stretch $A$ can produce.

*Why.* A unit vector $x = \sum_i c_i v_i$ (with $\sum c_i^2 = 1$) is sent to $Ax = \sum_i \sigma_i c_i u_i$. The $u_i$ are perpendicular, so the coefficient along $u_i$ is $\sigma_i c_i$, which can reach at most $\sigma_i$ (when $x = v_i$).

**Example 1: the filter.** For $A$ the SVD turns out to be (we compute it in Section 5) $\sigma_1 \approx 1.62$, $\sigma_2 = 1$, with $v_1 \approx (-0.78, 0.62)^T$, $v_2 \approx (-0.62, -0.78)^T$ in the canvas and $u_1 \approx (-0.79, 0.38, -0.48)^T$, $u_2 \approx (0, -0.78, -0.62)^T$, $u_3 \approx (-0.62, -0.49, 0.62)^T$ in 3D. So the unit circle on the canvas becomes an ellipse with semi-axes $1.62\,u_1$ and $1.0\,u_2$. Both $u_1$ and $u_2$ lie in the sheet $y_3 = y_1 + 0.8y_2$ of Section 1, so the ellipse lies in the sheet. The third direction $u_3$ is perpendicular to the sheet and is never reached (Figure 4). This is the extra zero row of $\Sigma$: after stretching, the picture sits in a plane inside $\mathbf{R}^3$, and $U$ rotates that plane onto the sheet.

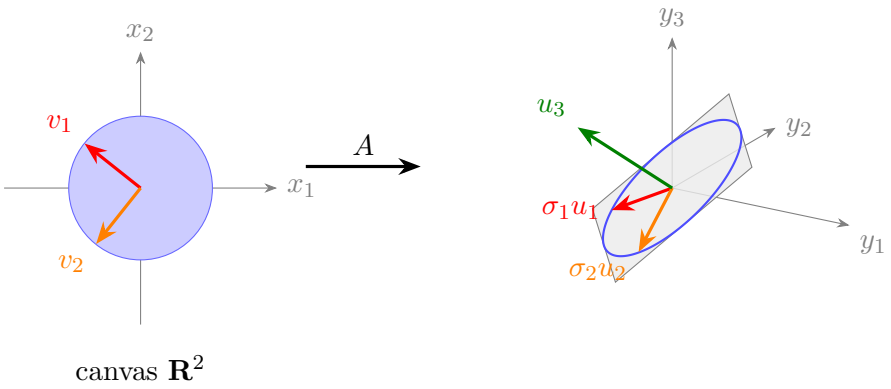



Figure 4: The circle becomes an ellipse in the sheet $\mathrm{col}(A)$. The semi-axes are $\sigma_1 u_1$ and $\sigma_2 u_2$ (red, orange), the images of $v_1$ and $v_2$. The direction $u_3$ (green) is perpendicular to the sheet and is never reached by $A$.

**Example 2: a $2 \times 2$ matrix.** Let $B = \left[\begin{smallmatrix} 3 & 0 \\ 4 & 5 \end{smallmatrix}\right]$, with

$$B = \underbrace{\frac{1}{\sqrt{10}}\begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} 3\sqrt{5} & 0 \\ 0 & \sqrt{5} \end{bmatrix}}_{\Sigma} \underbrace{\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}}_{V^T}$$

(computed in Section 5; you can already check $Bv_1 = 3\sqrt{5}\,u_1$ with $v_1 = \frac{1}{\sqrt{2}}(1,1)^T$ and $u_1 = \frac{1}{\sqrt{10}}(1,3)^T$). Here $V^T$ is a rotation by $-45°$, $\Sigma$ stretches the axes by $3\sqrt{5} \approx 6.71$ and $\sqrt{5} \approx 2.24$, and $U$ is a rotation by $+71.6°$. Figure 5 follows the circle around: across the top is $B$ in one step; the long way round is $V^T$, then $\Sigma$, then $U$, and both routes give the same ellipse. Its area is $\pi\sigma_1\sigma_2 = 15\pi$, so $B$ multiplies areas by $15 = |\det B|$.

## 4.2 What the matrix reaches, and what it destroys

If $r$ of the singular values are nonzero, the picture above tells us which parts of the input and output spaces matter:

- $u_1, \ldots, u_r$ span the **column space** $\mathrm{col}(A) \subseteq \mathbf{R}^m$, of dimension $r$: these are all the outputs that can occur;
- $v_{r+1}, \ldots, v_n$ span the **null space** $\mathrm{null}(A) \subseteq \mathbf{R}^n$, of dimension $n - r$: a zero singular value squashes that direction to nothing, so these inputs are sent to 0;
- $u_{r+1}, \ldots, u_m$ are never reached; they span $\mathrm{null}(A^T)$.

For the filter $r = 2 = n$, so nothing is destroyed, and the output is the 2D sheet inside $\mathbf{R}^3$. A matrix that *does* destroy something is $R = \left[\begin{smallmatrix} 1 & 2 \\ 2 & 4 \\ 2 & 4 \end{smallmatrix}\right]$ (second column $= 2\times$ first): its singular values are $\sigma_1 = 3\sqrt{5}$ and $\sigma_2 = 0$. Its ellipse has zero width, so the circle is flattened onto a *line segment* of length $2\sigma_1 \approx 13.4$, and the direction $v_2 = (-2, 1)^T/\sqrt{5}$ is lost for good: no one can undo the flattening.

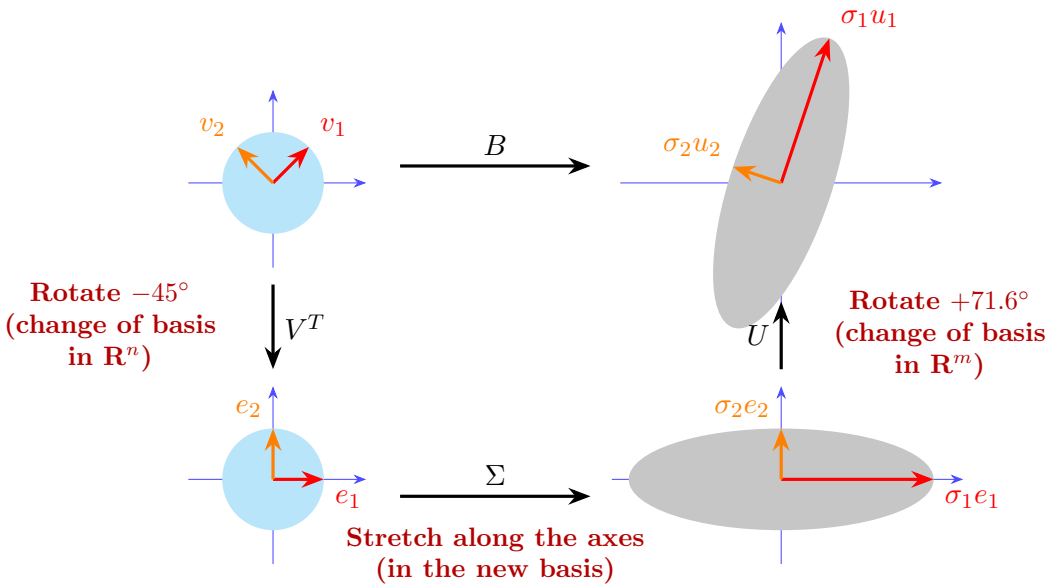



Figure 5: The SVD as a loop for $B$: rotate the perpendicular directions $v_1, v_2$ onto the axes, stretch the axes by $\sigma_1, \sigma_2$, rotate onto $u_1, u_2$. The circle becomes an ellipse whose axes are $\sigma_1 u_1$ and $\sigma_2 u_2$. Both routes give the same ellipse, which is what $B = U\Sigma V^T$ says.

**Symmetric matrices.** If $A$ is symmetric, then $\sigma_i = |\lambda_i|$ and $U$ and $V$ agree up to signs; a negative eigenvalue just becomes a flip inside $U$. (For $M$ of Lecture 7, $\lambda = 4, -1$ gives $\sigma = 4, 1$.) For a general matrix there is no such relation, which is why the SVD needs two sets of vectors.

**Check yourself.** What does $R$ do to the unit circle, and how long is the output?
*Answer.* It flattens it onto a segment along $u_1$, from $-\sigma_1 u_1$ to $\sigma_1 u_1$, of length $2\sigma_1 = 6\sqrt{5} \approx 13.4$.

# 5 Computing the SVD

## 5.1 Which directions get stretched most?

We know what we are looking for: perpendicular input directions $v_i$ that $A$ stretches by $\sigma_i$. How do we find them? Here the observation of Section 2.3 pays off: the stretch of a unit vector $x$ is $\|Ax\|$, and

$$\|Ax\|^2 = x^T(A^TA)x = \sum_i \lambda_i c_i^2 \qquad (c = \text{coordinates of } x \text{ in the eigenvector basis of } A^TA).$$

**In words:** the squared stretch is a weighted average of the eigenvalues of $A^TA$ (the weights $c_i^2$ add up to 1 for a unit vector). It is largest when we put all the weight on the biggest eigenvalue, that is, when $x$ is the top eigenvector of $A^TA$, and the largest stretch is $\sqrt{\lambda_1}$. The next best perpendicular direction is the second eigenvector, and so on. Therefore:

> **Key fact.** The right singular vectors $v_i$ are the eigenvectors of $A^TA$, and the singular values are $\sigma_i = \sqrt{\lambda_i}$, where $\lambda_i \geq 0$ are its eigenvalues.

The same fact in matrix form: if $A = U\Sigma V^T$ then $A^TA = V\,(\Sigma^T\Sigma)\,V^T$, an eigen-decomposition with eigenvalues $\sigma_i^2$. (Since $A^TA$ is PSD the square roots exist, which is why singular values are never negative.)

### 5.2 Finding the left singular vectors

We know $Av_i = \sigma_i u_i$. So once $v_i$ and $\sigma_i$ are known, the left singular vector is determined: it is the image $Av_i$, divided by its length $\sigma_i$ so that it becomes a unit vector,

$$u_i = \frac{Av_i}{\sigma_i} \qquad (\sigma_i > 0).$$

These $u_i$ are automatically perpendicular to one another: $u_i^T u_j = \dfrac{v_i^T (A^T A v_j)}{\sigma_i \sigma_j} = \dfrac{\lambda_j\, v_i^T v_j}{\sigma_i \sigma_j} = 0$ for $i \neq j$, because the $v_i$ are perpendicular. If $m > r$, the missing columns of $U$ are chosen as any unit vectors perpendicular to $u_1, \ldots, u_r$ (they are the directions that $A$ never reaches). This construction works for every matrix, so it also proves that *every matrix has an SVD.* (The matrices $A^T A$ and $AA^T$ have the same nonzero eigenvalues, since $A^T Av = \lambda v$ implies $AA^T(Av) = \lambda(Av)$; the larger one just has extra zeros.)

**Warning about signs.** It is tempting to take $U$ from a separate eigen-decomposition of $AA^T$. Do not. An eigenvector is only defined up to sign, and only one pairing of $u_i$ with $v_i$ satisfies $Av_i = \sigma_i u_i$. Always get $V$ from $A^T A$ and then **define** $u_i = Av_i/\sigma_i$.

**Recipe.**
1. Form $A^T A$ and find its eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq 0$. Then $\sigma_i = \sqrt{\lambda_i}$.
2. Find unit eigenvectors $v_i$: these are the columns of $V$.
3. For each $\sigma_i > 0$ set $u_i = Av_i/\sigma_i$.
4. If $m > r$, complete $U$ with unit vectors perpendicular to $u_1, \ldots, u_r$ (they solve $A^T w = 0$).

**Sanity checks:** $\sum \sigma_i^2 =$ sum of squares of all entries of $A$; for square $A$, $\sigma_1 \cdots \sigma_n = |\det A|$.

**Example 3: $B = \left[\begin{smallmatrix} 3 & 0 \\ 4 & 5 \end{smallmatrix}\right]$ by hand. Step 1** (the matrix that measures stretching): $B^T B = \left[\begin{smallmatrix} 25 & 20 \\ 20 & 25 \end{smallmatrix}\right]$. **Step 2** (stretch factors): its characteristic polynomial is $(25-\lambda)^2 - 400 = (\lambda - 45)(\lambda - 5)$, so

$$\sigma_1 = \sqrt{45} = 3\sqrt{5} \approx 6.71, \qquad \sigma_2 = \sqrt{5} \approx 2.24.$$

(Check: $\sigma_1 \sigma_2 = 15 = |\det B|$.) **Step 3** (stretched directions): for $\lambda = 45$, $(B^T B - 45I)v = 0$ says the two entries of $v$ are equal, so $v_1 = \frac{1}{\sqrt{2}}(1,1)^T$; for $\lambda = 5$ the entries are opposite, so $v_2 = \frac{1}{\sqrt{2}}(-1,1)^T$. **Step 4** (where they land):

$$u_1 = \frac{Bv_1}{\sigma_1} = \frac{1}{\sqrt{10}}\begin{bmatrix} 1 \\ 3 \end{bmatrix}, \qquad u_2 = \frac{Bv_2}{\sigma_2} = \frac{1}{\sqrt{10}}\begin{bmatrix} -3 \\ 1 \end{bmatrix}.$$

Putting the pieces together gives the factorisation of Example 2. *The sign trap:* $BB^T$ also has $\frac{1}{\sqrt{10}}(3,-1)^T$ as an eigenvector for $\sigma_2$, but using it in place of $u_2$ produces $\left[\begin{smallmatrix} 0 & 3 \\ 5 & 4 \end{smallmatrix}\right] \neq B$. Notice also that the eigenvectors of $B$ itself ($\lambda = 3, 5$) are not perpendicular and have nothing to do with $\sigma_1, \sigma_2$: SVD is not the same as eigen-decomposition.

**Example 4: the lifting filter. Step 1:** $A^T A = \left[\begin{smallmatrix} 2 & -0.8 \\ -0.8 & 1.64 \end{smallmatrix}\right]$. **Step 2:** trace 3.64, determinant 2.64, so $\lambda^2 - 3.64\lambda + 2.64 = (\lambda - 2.64)(\lambda - 1)$ and

$$\sigma_1 = \sqrt{2.64} \approx 1.62, \qquad \sigma_2 = 1.$$

**Step 3:** the unit eigenvectors are $v_1 \approx (-0.78,\, 0.62)^T$ and $v_2 \approx (-0.62,\, -0.78)^T$ (perpendicular, as they must be). **Step 4:** $u_i = Av_i/\sigma_i$ gives $u_1 \approx (-0.79,\, 0.38,\, -0.48)^T$ and $u_2 \approx$

$(0,\,-0.78,\,-0.62)^T$. Since $m = 3 > r = 2$, one more column is needed, a unit vector perpendicular to both. It must solve $A^T w = 0$, which gives $u_3 = \frac{1}{\sqrt{2.64}}(-1, -0.8, 1)^T \approx (-0.62, -0.49, 0.62)^T$: the normal of the sheet. So

$$A = \underbrace{\begin{bmatrix} -0.79 & 0 & -0.62 \\ 0.38 & -0.78 & -0.49 \\ -0.48 & -0.62 & 0.62 \end{bmatrix}}_{U\ (3\times3)} \underbrace{\begin{bmatrix} 1.62 & 0 \\ 0 & 1.0 \\ 0 & 0 \end{bmatrix}}_{\Sigma\ (3\times2)} \underbrace{\begin{bmatrix} -0.78 & 0.62 \\ -0.62 & -0.78 \end{bmatrix}}_{V^T\ (2\times2)},$$

the decomposition on the slides. (Check: $Av_1 = (-1.28,\, 0.62,\, -0.78)^T = 1.62\, u_1$.)

**Check yourself.** What is the SVD of $D = \mathrm{diag}(3, -2)$?
*Answer.* Singular values 3 and 2 (they cannot be negative). The minus sign moves into $U$: $D = \mathrm{diag}(1, -1)\ \mathrm{diag}(3, 2)\, I$. A flip is a rotation-like step, so it belongs in $U$, not in the stretch.

# 6 Rank and rank estimation

The *rank* of a matrix is the number of dimensions its output really uses. The ellipse picture makes this easy to read off: each nonzero singular value is an axis of the ellipse with positive length, and each zero singular value is a direction squashed to nothing.

**Key fact.** The rank of $A$ equals the number of nonzero singular values.

*Why.* In $A = U\Sigma V^T$ both $U$ and $V^T$ are invertible (rotations lose no information), and multiplying by an invertible matrix does not change the rank, so $\mathrm{rank}\, A = \mathrm{rank}\, \Sigma$, which counts the nonzero diagonal entries. (The slides say "rank = number of nonzero eigenvalues". That is true for symmetric matrices such as $A^T A$, but not for every square matrix: $\left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$ has rank 1 and both eigenvalues 0.)

## 6.1 Application: the rank of noisy data

Here is a practical use. Suppose we measure $n$ points in $\mathbf{R}^m$ and stack them as the columns of $A \in \mathbf{R}^{m\times n}$. If the points really lie on an $r$-dimensional subspace (a line, a plane, ...), the rank of $A$ is $r$, and we would like to discover $r$. But measurements are noisy, so $A$ has full rank in practice. The SVD still sees the structure: *$r$ singular values are large, and the others are only as large as the noise.* Counting the large ones gives the "approximate rank".

**Example 5.** Eight points in $\mathbf{R}^3$ lie on the plane $y_3 = y_1 + y_2$ (so the true rank is 2), but each entry carries noise of size about 0.05:

$$Y = \begin{bmatrix} 0.43 & 1.58 & 1.00 & -1.16 & -0.89 & 1.49 & -2.06 & 1.31 \\ 1.21 & -0.11 & -0.93 & -0.93 & -1.00 & -0.19 & -0.08 & 0.18 \\ 1.65 & 1.46 & 0.35 & -2.04 & -1.80 & 1.34 & -2.03 & 1.49 \end{bmatrix} \in \mathbf{R}^{3\times8}.$$

The singular values are $\sigma_1 \approx 5.83$, $\sigma_2 \approx 2.20$, $\sigma_3 \approx 0.15$ (Figure 6). Two are large and one is tiny, so the data have approximate rank 2, *without us ever being told about the plane.* Geometrically, $u_3 \approx (-0.60, -0.54, 0.59)^T$ is almost the normal $(-1, -1, 1)/\sqrt{3}$ of the true plane: it is the direction in which the data barely vary. There is no universal cutoff; one looks for a clear gap ($2.20 \to 0.15$ here), because with noise no singular value ever comes out exactly zero.

**Check yourself.** A $4 \times 100$ data matrix has singular values 10, 9.5, 0.01, 0.008. What is its approximate rank?
*Answer.* 2: the 100 points in $\mathbf{R}^4$ lie almost exactly on a 2-dimensional subspace.

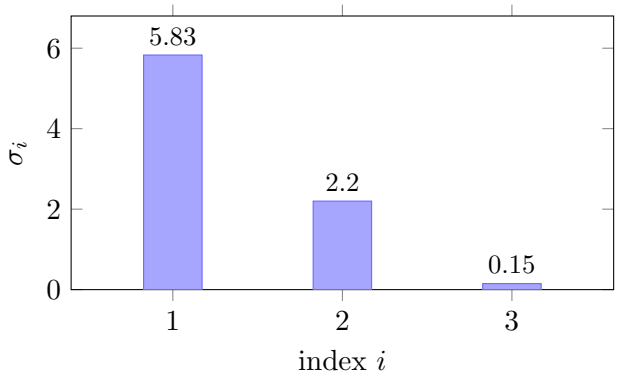



Figure 6: Singular values of the noisy data matrix $Y$. Two are large and one is tiny: approximate rank 2.

# 7 Matrix approximation

## 7.1 A matrix as a sum of layers

The singular values are ordered by importance, which suggests a way of simplifying a matrix: keep the important parts and throw away the rest. To make this precise, multiply out $A = U\Sigma V^T$ one column at a time:

$$\boxed{A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \cdots + \sigma_r u_r v_r^T}$$

**In words:** $A$ is a sum of simple "layers". Each layer $u_i v_i^T$ is an $m \times n$ matrix of rank one (a column times a row): one pattern in the output space tied to one pattern in the input space, with strength $\sigma_i$. The first layer is the strongest, and each later layer adds a finer correction. Think of a photograph built up from a coarse outline first, then more and more detail.

Keeping only the first $k$ layers gives the **$k$-rank approximation**

$$A_k = \sigma_1 u_1 v_1^T + \cdots + \sigma_k u_k v_k^T, \qquad \operatorname{rank} A_k = k.$$

How good is it? The error is the sum of the layers we threw away, so it is controlled by the singular values we threw away: the largest stretch of the error matrix $A - A_k$ is $\sigma_{k+1}$, and the total squared error (the sum of the squares of all entries of $A - A_k$) is $\sigma_{k+1}^2 + \sigma_{k+2}^2 + \cdots + \sigma_r^2$. **In words:** if the discarded singular values are small, $A_k$ is almost $A$. (A standard theorem of Eckart and Young says that no other rank-$k$ matrix does better; we use it without proof.) The fraction of the "energy" kept is $\sum_{i \le k} \sigma_i^2 / \sum_i \sigma_i^2$.

**Example 6: $B$.** From Example 3, $B_1 = \sigma_1 u_1 v_1^T = \frac{3}{2}\left[\begin{smallmatrix} \frac{1}{3} & \frac{1}{3} \end{smallmatrix}\right]$ and $B - B_1 = \sigma_2 u_2 v_2^T = \frac{1}{2}\left[\begin{smallmatrix} 3 & -3 \\ -1 & 1 \end{smallmatrix}\right]$. The total squared error is $2.25 + 2.25 + 0.25 + 0.25 = 5 = \sigma_2^2$, and the energy kept is $45/50 = 90\%$. One layer already captures 90% of $B$.

**Example 7: the filter.** $A_1 = (Av_1)v_1^T \approx \left[\begin{smallmatrix} 1 & -0.8 \\ -0.49 & 0.39 \\ 0.61 & -0.49 \end{smallmatrix}\right]$, the largest error stretch is $\sigma_2 = 1$, and the energy kept is $2.64/3.64 \approx 72.5\%$. Here the second singular value is not small, so one layer is a poor approximation.

## 7.2 The geometry of dropping a layer

Dropping the layer $\sigma_2 u_2 v_2^T$ removes the *short axis* of the ellipse. The approximation $A_1$ sends the circle not to an ellipse but to the *segment* along the long axis (Figure 7). What is lost is the half-width $\sigma_2$ of the ellipse. The thinner the ellipse already is (the smaller $\sigma_2$), the better a line approximates it.

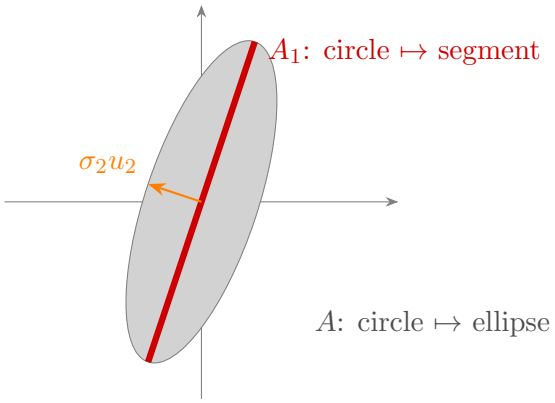



Figure 7: Keeping one layer. The ellipse (all of $A$) and the segment along its long axis (the rank-one approximation $A_1$). What is lost is the half-width $\sigma_2 u_2$.

### 7.3 Why it matters: compression

An $m \times n$ matrix stores $mn$ numbers. The $k$-rank approximation needs only $k$ vectors $u_i \in \mathbf{R}^m$, $k$ vectors $v_i \in \mathbf{R}^n$ and $k$ numbers $\sigma_i$:

$$\text{storage}(A_k) = k(m + n + 1).$$

For a grey-scale image of $1000 \times 800$ pixels and $k = 50$ this is $50 \cdot 1801 = 90{,}050$ numbers instead of 800,000, about 11%. Images are approximately low rank, so their singular values fall off quickly and a small $k$ keeps almost all of the energy. Choosing $k$ is the same decision as estimating the approximate rank in Section 6.1: look for where the singular values drop.

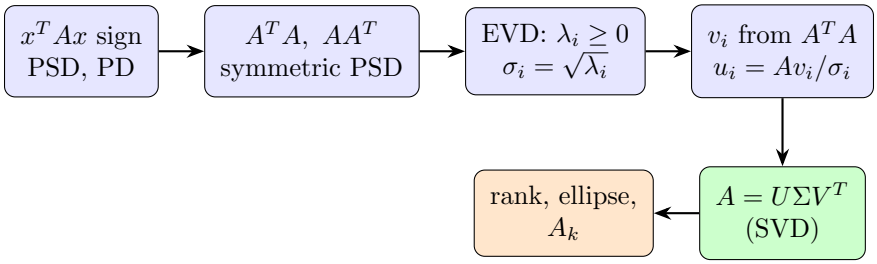
**Check yourself.** A matrix has singular values $4, 3, 0, 0$. What is the energy kept by $A_1$, and what is $A_2$?
*Answer.* $16/(16 + 9) = 64\%$. Since $r = 2$, $A_2 = A$ exactly: there is nothing more to keep.

**In NumPy.** The calculations of this lecture take a few lines:

```
import numpy as np
A = np.array([[1, -0.8], [0, 1], [1, 0]])
U, s, Vt = np.linalg.svd(A)              # U: 3x3, s: 2 values, Vt: 2x2
print(s)                                 # [1.6248 1.    ]
A1 = s[0] * np.outer(U[:, 0], Vt[0])     # rank-1 approximation
print(np.linalg.norm(A - A1, 2))         # 1.0 = sigma_2
```

(NumPy may flip the signs of a pair $u_i, v_i$; the product is the same.)

## 8 Summary map



| **Idea** | **Key fact** | **In the examples** |
|---|---|---|
| Definite matrices | Symmetric: sign of $x^TAx$ follows the signs of the eigenvalues; $x^TA^TAx = \|Ax\|^2$, so $A^TA$, $AA^T$ are always PSD | $S$ PD, $F$ PSD, $M$ indefinite |
| SVD | $A = U\Sigma V^T$, $Av_i = \sigma_i u_i$ | Shapes $3\times3$, $3\times2$, $2\times2$ for the filter |
| Geometry | Rotate, stretch, rotate: circle $\to$ ellipse with semi-axes $\sigma_i u_i$ | Circle $\to$ ellipse on the sheet; rank-1 $R$: circle $\to$ segment |
| Computing | $v_i$ eigenvectors of $A^TA$, $\sigma_i = \sqrt{\lambda_i}$, $u_i = Av_i/\sigma_i$ (never two separate EVDs) | $\sigma = 1.62$, 1 (filter); $3\sqrt{5}, \sqrt{5}$ ($B$) |
| Rank | $\operatorname{rank} A = \#\{\sigma_i > 0\}$; approximate rank from the gap | $Y$: 5.83, 2.20, 0.15 $\Rightarrow$ rank 2 |
| Approximation | $A_k = \sum_{i\le k} \sigma_i u_i v_i^T$; error from the discarded $\sigma_i$; storage $k(m+n+1)$ | $B_1$: 90% energy kept |

**SVD versus EVD.** The eigen-decomposition needs a square matrix and may need complex numbers. The SVD exists for every real matrix of any shape, always uses real orthogonal $U, V$, and has non-negative $\sigma_i$. For symmetric matrices the two agree up to signs: $\sigma_i = |\lambda_i|$.